

SHUBHAM KUMAR

+91-9588885199

shubhamdevu023@gmail.com

LinkedIn

GitHub

Portfolio

Professional Summary

Software Engineer with 1+ years of experience in ReactJS, TypeScript, and real-time systems — from shipping enterprise video platforms at SnetLabs to publishing a React component library on npm. Currently building **JustSay**, a production AI SaaS that generates and runs full React apps live in the browser.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL

Frontend: ReactJS, Next.js, Redux Toolkit, Context API, HTML5, CSS3, Tailwind CSS, Framer Motion, GSAP

Backend & Infra: Node.js, Express.js, REST APIs, WebSockets, MQTT, PostgreSQL, Docker, AWS EC2, GCP, JWT, OAuth

AI & Tools: OpenAI API, LLM Orchestration, WebContainers (StackBlitz), SSE Streaming, Prompt Engineering, Git, Vercel

Practices: Agile, Component-Driven Development, Viewport Virtualisation, Performance Optimisation, Code Reviews

Work Experience

SnetLabs

Software Engineer

Dec 2024 – Present

Noida, UP

- Co-architected a browser-side control plane for an enterprise VMS using a **custom playback engine driven by requestAnimationFrame**, enabling frame-accurate sync across **100+ concurrent camera streams**.
- Replaced polling with a **WebSocket/MQTT signalling layer**, achieving **sub-50ms latency** for PTZ camera controls and instant AI alert delivery — eliminating a core reliability bottleneck in the platform.
- Implemented **Viewport Virtualisation** with scroll-throttled subscribe/unsubscribe signals, cutting stream signalling load proportionally to off-screen feeds and keeping the UI fluid under heavy grid layouts.
- Partitioned React Context and batched state updates to **reduce re-renders by 40–50%**, sustaining a consistent **60 FPS** experience across high-density, data-heavy dashboard views.
- Built a **React Canvas visual workflow editor** for composing AI analytics pipelines graphically, enabling non-technical operators to configure and deploy complex worker processes without engineering support.
- Co-authored **labsense-ui-kit** (npm) — a shared React + TypeScript component library that standardised design patterns across teams and **accelerated feature delivery by 25%**.

Potterstech

Frontend Engineer Intern

June 2023 – Aug 2023

Remote

- Built a library of **responsive, reusable ReactJS + Tailwind CSS components** for a cross-browser production application, shipped to paying customers within the internship window.
- Integrated **10+ RESTful API endpoints** with full async lifecycle handling — loading skeletons, error boundaries, and empty states — significantly improving perceived performance and UX consistency.
- Implemented **interactive UI features** using React Hooks (useState, useEffect, useCallback) and modern JS patterns, reducing boilerplate and improving render efficiency across feature modules.
- Collaborated with senior engineers to **refactor 3 legacy modules**, consolidating duplicated logic into shared utilities and improving long-term maintainability of the frontend codebase.

Featured Projects

JustSay - AI Website builder TypeScript, React, Node.js, WebContainers, OpenAI API, PostgreSQL, AWS

- Building a **production SaaS** that converts natural language prompts into fully functional Vite + React apps, executed live inside **WebContainers** (StackBlitz) — zero server-side runtime, zero file system access, instant in-browser preview.
- Engineered a **7-stage LLM orchestration pipeline** (classify → expand → plan → skeleton → batch-generate → normalise → fix) with all output streamed token-by-token via **SSE**, giving users real-time file materialisation feedback.
- Designed a **two-tier caching system** — exact-prompt fingerprint cache + semantic category cache (S3) — intercepting redundant LLM calls before inference, slashing API credit spend and **p95 generation latency** for repeated categories.
- Architecting a **multi-tenant container orchestration layer** with per-user WebContainer lifecycle management: spawn, idle detection, graceful teardown, and resource quotas to enforce fairness under concurrent load.
- Implementing **follow-up prompt chaining** with full project-context persistence so users can iteratively refine generated sites conversationally — each edit costs a delta, not a full regeneration.

WeTube — Full-Stack Video Platform

MERN, Firebase, Redux Toolkit, JWT

- Architected a **full-stack YouTube-style platform** end-to-end — Express + MongoDB backend with RESTful APIs for video CRUD, subscriptions, likes, comments, and channel management.
- Integrated **Firebase Storage** for chunked video and thumbnail uploads with upload progress tracking, and secured all routes with **JWT + Google OAuth** authentication.
- Implemented **Redux Toolkit** global state with normalised entity slices for videos, users, and subscriptions, eliminating redundant API calls via client-side cache invalidation.
- Deployed on **Vercel** (frontend) and **Render** (backend) with environment-based config, responsive UI across all breakpoints, and optimised asset loading for core video pages.

TSP via Differential Evolution

Python, NumPy, Matplotlib

- Designed a **Differential Evolution** solver for the Travelling Salesman Problem with custom mutation (DE/rand/1) and crossover strategies, achieving high-quality route optimisation on standard benchmark datasets.
- Outperformed traditional **Genetic Algorithm** baselines on convergence speed and solution quality — benchmarked across multiple city-count configurations with consistent tour-length improvements.
- Implemented **population diversity controls** and adaptive crossover rate tuning to prevent premature convergence, enabling the solver to escape local optima on high-complexity instances.

Education

University of Petroleum and Energy Studies

B.Tech in Computer Science — 7.6 CGPA

2020 – 2024

Dehradun, Uttarakhand